Gaussian Process Regression

Predicting Building Heating and Cooling Load from Shape Features

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Abstract. Building energy performance prediction is critical for sustainable design, yet physics-based simulations remain computationally expensive. This study evaluates Gaussian Process Regression (GPR) with a Matérn-1.5 kernel as a probabilistic surrogate for predicting heating load (Y1) and cooling load (Y2) from the UCI Energy Efficiency dataset (768 samples, 8 features). The Gaussian-process model is benchmarked against ridge regression and random forests within a reproducible scikit-learn pipeline that standardizes continuous features and one-hot encodes categorical design variables. Evaluation covers accuracy (MAE, RMSE, R^2), predictive interval coverage at 90% and 95% levels, and standardized residual diagnostics. Results show that the probabilistic surrogate achieves competitive accuracy with the random forest baseline while uniquely providing model-based predictive uncertainty. Empirical coverage is strong for Y1 and mildly underconfident for Y2, and a post-hoc split conformal check reduces that gap. The study highlights a practical trade-off between point accuracy and reliable confidence intervals for design screening.

I. Introduction

Predicting building energy demand requires fast surrogates because physics-based simulations are expensive [1], [2], [3]. Gaussian Process Regression (GPR) is attractive in this setting because it combines strong tabular regression with model-based uncertainty that can support design screening and risk flagging [4], [5]. In early-stage design, those uncertainty estimates help rank candidate designs and flag risky regions before expensive follow-up analysis. Recent surveys likewise stress that transparent uncertainty estimates matter once models start informing design decisions [2], [3]. This study asks whether a Gaussian-process model can predict heating and cooling load accurately while providing useful uncertainty estimates from basic building features. The objective is therefore not only low point error, but also an uncertainty signal that can be checked against held-out outcomes. The study contributes a reproducible benchmark with principled categorical handling, a numeric-encoding ablation, and interval-based uncertainty diagnostics under a fixed train/test split.

II. Data

The Energy Efficiency dataset contains 768 simulated buildings with 8 features and 2 targets generated in Ecotect by varying glazing area, glazing distribution, and orientation [1]. There are no missing values; X6 (Orientation, 4 levels) and X8 (Glazing Area Distribution, 6 levels) are treated as categorical, while the remaining predictors take discrete numeric design levels. Heating load (Y1) ranges from 6.0 to 43.1 and cooling load (Y2) from 10.9 to 48.0. Although simulation-based, the dataset forms a controlled design grid and remains a standard benchmark for surrogate modeling in this setting [1]. The two targets are strongly correlated, but they are modeled separately here to keep target-wise residual and coverage checks interpretable. The dataset therefore forms a structured design grid rather than an unconstrained continuous sample, so encoding choices can change both point accuracy and how uncertainty summaries should be read. Table I defines the variables used below, while Figs. 1 and 2 summarize the marginal and joint structure most relevant for modeling.

Table I. Feature definitions from the Energy Efficiency dataset [1].

Feature	Description
X1	Relative compactness
X2	Surface area
X3	Wall area
X4	Roof area
X5	Overall height
X6	Orientation (categorical)
X7	Glazing area
X8	Glazing area distribution (categorical)
Y1	Heating load
Y2	Cooling load

Source: Author's elaboration based on [1].

Fig. 1 shows repeated design levels rather than smooth marginals. That structure motivates one-hot encoding for X6/X8 and cautions against reading uncertainty summaries as if the inputs were sampled from a continuous population. It also helps explain why Y1 and Y2 are modeled separately downstream: both targets span broader and less regular ranges than several inputs, making target-wise residual and coverage checks easier to interpret. Category counts are reported in Appendix Fig. A1, where they primarily confirm balance across the categorical levels used in the main analysis.

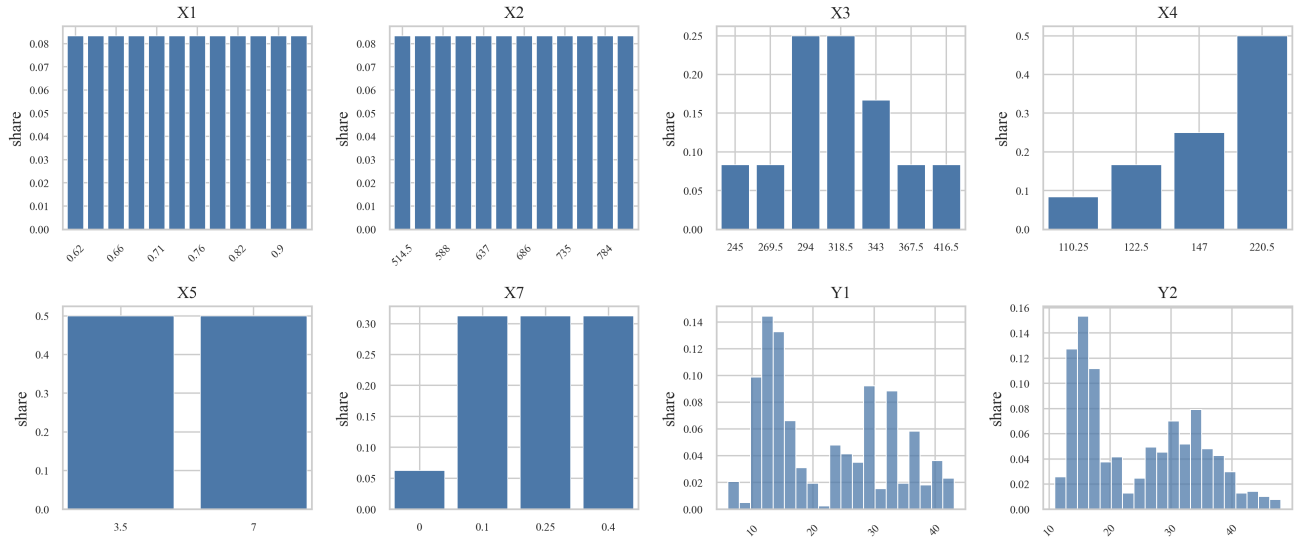


Fig. 1. Distributions for numeric features (X1 to X5 and X7) and targets (Y1 and Y2); X6 and X8 are categorical and summarized separately in Appendix Fig. A1. Discrete feature levels are shown as shares per level.

Source: Author's elaboration.

To move from marginal distributions to the joint target structure, Fig. 2 uses X8 because glazing distribution provides the clearest visible partition of the Y1/Y2 space.

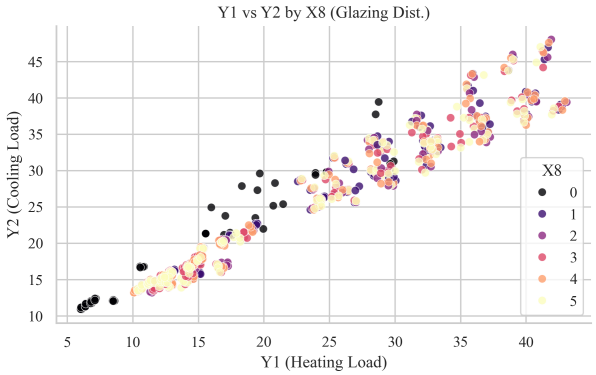


Fig. 2. Joint relationship between Y1 and Y2 colored by glazing distribution (X8). Across the full dataset, the target correlation is strong ($r \approx 0.98$).

Source: Author's elaboration.

The joint Y1/Y2 trend remains tight, but the X8 coloring separates it into visible strata. Glazing distribution yields the clearest visual partition here, so the orientation-colored companion view is left in Appendix Fig. A1. This supports target-wise models with shared preprocessing, while residual and calibration diagnostics are deferred to the appendix.

III. Methods

A. Models

Ridge regression, random forest regression, and Gaussian Process Regression are implemented in scikit-learn [6]; for a tabular dataset of moderate size and discrete design levels, these baselines provide a sensible comparison [7]. Ridge regression provides a linear reference [8], random forests capture nonlinear interactions [9], and the Gaussian-process model supplies closed-form predictive uncertainty [4], [5]. The GP uses a Matérn kernel with a white-noise term: $k(x, x') = \sigma^2 \cdot \text{Matérn}_{\nu=1.5}(\ell) + \sigma_n^2 I$. Hyperparameters are optimized by log-marginal likelihood with a small number of restarts to keep CPU runtime manageable. Matérn-$\nu = 1.5$ is less smooth than the squared-exponential kernel and better matched to the piecewise design levels present here, while the white-noise term absorbs simulation artifacts and residual lack-of-fit [10], [4].

B. Uncertainty and Calibration

The Gaussian-process model yields a predictive mean $\mu(x)$ and standard deviation $\sigma(x)$ from the GP posterior, with the white-noise term keeping the uncertainty discussion explicit in terms of epistemic model variance and irreducible simulation artifacts or lack-of-fit [4], [11]. Initial uncertainty is assessed through standardized residuals $z = (y - \mu)/\sigma$, which should be approximately $\mathcal{N}(0, 1)$ for a well-behaved model. To evaluate the empirical validity of the 90% and 95% predictive intervals, coverage is reported in the spirit of conformal-prediction diagnostics [12]. Reliability curves compare nominal and empirical coverage across multiple confidence levels and are treated here as empirical checks rather than proof of calibration.

IV. Model Building

A single scikit-learn pipeline uses StandardScaler for continuous features (X1 to X5 and X7) and OneHotEncoder for categorical features (X6, X8). Separate models are trained per target (Y1, Y2) to keep uncertainty

analysis interpretable. Baselines are RidgeCV with 5-fold training-set CV and a 300-tree RandomForestRegressor under the same outer split and preprocessing pipeline. The random forest keeps the default tree depth and uses 300 trees as a stable untuned baseline that captures nonlinear interactions without an extra tuning loop. The GP pipeline uses a Constant × Matérn kernel plus white noise with $n_restarts_optimizer = 2$, which keeps CPU runtime manageable at $n = 768$. Matérn-$\nu = 1.5$ balances smoothness and flexibility for the piecewise design levels present in the dataset. A numeric ablation treats X6/X8 as ordinal inputs (GPR Numeric) under the same split and diagnostics.

V. Evaluation

An 80/20 split with fixed seed (42) is reused across models and targets for direct comparison. Repeated K-fold validation was bypassed in the main text to preserve a single, unconfounded calibration protocol for the post-hoc conformal coverage check, while full 10-split stability results are reported in Appendix Table A1. Held-out metrics are MAE, RMSE, and R^2, defined in Eqs. (1) to (3); for the Gaussian-process model, interval coverage is derived from predictive standard deviations and Gaussian quantiles, while reliability curves summarize nominal coverage levels.

$$\mathrm{MAE} = \frac{1}{n}\sum_{i=1}^{n}|y_i - \hat{y}_i|, \tag{1}$$

$$\mathrm{RMSE} = \sqrt{\frac{1}{n}\sum_{i=1}^{n}(y_i - \hat{y}_i)^2}, \tag{2}$$

$$R^2 = 1 - \frac{\sum_{i=1}^{n}(y_i - \hat{y}_i)^2}{\sum_{i=1}^{n}(y_i - \bar{y})^2}. \tag{3}$$

Standardized residuals use $z_i = (y_i - \mu_i)/\sigma_i$, and predictive interval coverage is

$$\mathrm{Cov}_\alpha = \frac{1}{n}\sum_{i=1}^{n}\mathbf{1}\{|y_i - \mu_i| \le z_\alpha \sigma_i\}, \tag{4}$$

with $z_{0.90} = 1.645$ and $z_{0.95} = 1.96$; all seeds and split indices are saved to ensure reproducibility.

VI. Results

Tables II and III report the single-split test metrics, while Appendix Table A1 shows that the ranking remains broadly stable across 10 repeated splits. Appendix Fig. A1 reports category balance and the orientation-colored companion view, and Appendix Figs. A2–A3 provide residual and uncertainty diagnostics. Coverage is conservative for Y1 and slightly below nominal for Y2 at the 95% level, with heavier Y2 tails in the standardized residuals (Appendix Fig. A3).

Table II. Test metrics for Y1 (Heating Load). Best values per metric are shown in bold; for $\mathrm{Cov}_{90}/\mathrm{Cov}_{95}$ the value closest to nominal coverage is bolded.

Model	MAE	RMSE	R^2	Cov_{90}	Cov_{95}
Ridge	2.061	2.874	0.921	n/a	n/a
Random Forest	**0.367**	**0.533**	**0.997**	n/a	n/a
GPR	0.518	0.703	0.995	0.961	**0.987**
GPR (Numeric)	0.433	0.578	0.997	**0.948**	**0.987**

Source: Author's elaboration.

Table III. Test metrics for Y2 (Cooling Load). Best values per metric are shown in bold; for $\mathrm{Cov}_{90}/\mathrm{Cov}_{95}$ the value closest to nominal coverage is bolded.

Model	MAE	RMSE	R^2	Cov_{90}	Cov_{95}
Ridge	2.180	3.120	0.895	n/a	n/a
Random Forest	1.270	2.035	0.955	n/a	n/a
GPR	1.038	1.497	0.976	**0.916**	0.929
GPR (Numeric)	**0.787**	**1.213**	**0.984**	0.935	**0.942**

Source: Author's elaboration.

Across both targets, Tables II and III show the same pattern: random forests minimize Y1 point error, while GP variants are stronger on Y2 and are the only baselines with interval estimates.

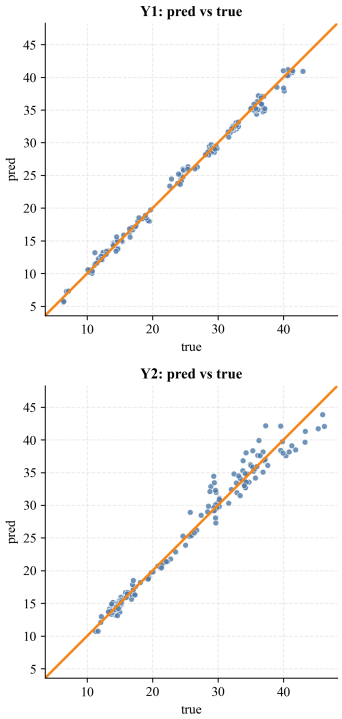


Fig. 3. Gaussian-process predictions versus true values for Y1 (Heating Load) and Y2 (Cooling Load).

Source: Author's elaboration.

The prediction plots in Fig. 3 confirm the ranking, with tighter Y1 alignment and broader Y2 spread. Random forests still yield the lowest Y1 errors, while the Gaussian-process variants remain competitive and are the only baselines in this comparison that yield interval estimates. The numeric ablation improves raw error but imposes an unjustified ordinal structure on X6/X8, and ridge underfits the nonlinear design relationships.

VII. Discussion

The numeric ablation improves point metrics only by imposing an unjustified ordinal distance, so one-hot encoding remains the principled choice. The GP surrogate is most useful for design screening because it remains competitive while exposing uncertainty that tree baselines lack. Our random-forest baselines achieve mean absolute errors of 0.37 for Y1 and 1.27 for Y2, consistent with the original Energy Efficiency study (0.51 and 1.42, respectively), although direct comparison is limited by split and preprocessing differences [1].

Key limitations are the fixed split, limited hyperparameter search, mild Y2 undercoverage, and the absence of proper scoring rules such as NLL or CRPS. Because the dataset is simulated and discrete, external validity is limited to the observed design space, although most standardized residuals remain within about ± 2. Appendix Table A2 shows that split conformal calibration [13] moves Y2 coverage close to nominal without changing the model class; future work can add richer kernels and validation. Given the extreme Y1/Y2 correlation ($r \approx 0.98$), future iterations should explore multi-output Gaussian processes to share strength across the two targets [4].

VIII. Conclusion

A Gaussian-process model with careful preprocessing gives strong heating and cooling load prediction with good Y1 coverage and mild Y2 undercoverage. Random forests deliver the lowest Y1 point error, whereas the GP surrogate provides an auditable uncertainty signal that can be recalibrated when needed. Ridge and numeric-encoding comparisons show that the best point metrics need not align with the most defensible uncertainty treatment.

AI Usage Statement

OpenAI ChatGPT and Anthropic Claude Code were used for code review, refactoring, and minor edits to the report text. The core codebase, experiments, and results were authored and executed by the student; AI was not used to generate results or citations.

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